

Income Taxation

11.1 INTRODUCTION

The ultimate objective of any approach to welfare economics is to provide criteria for the evaluation of social policies. In this chapter, we show how the social ordering functions axiomatized in the previous chapter engender such criteria.

After a policymaker has chosen the social ordering function that captures the policymaker's her ethical preferences, he or she typically faces information constraints that prevent him or her from being able to implement the allocations that are optimal according to the chosen social ordering function. The information constraints may be of different sorts, and we show here how the social ordering functions we have defined adjust to these different informational structures.

We study two informational structures. In some applications, we assume that labor time and earnings are observable, but skills are not. Agents may then choose to work at a lower wage rate than their skill warrants, if it is in their interest to do so. In other applications, we assume that labor time is no longer observable, and only earnings are. Under either set of assumptions, the policymaker must identify the optimal allocation among the ones that are compatible with the agents' incentives to hide their private information.

Our social ordering function approach turns out to accommodate these different structures easily. This comes from two central properties, on which we have already insisted. The first property is that the social ordering functions that satisfy the axioms we impose are of the leximin type. Consequently, social welfare is appropriately measured by the well-being index of the worst-off agents. It turns out to be measured easily. The second property is the informational simplicity of the well-being indices we use. They rely on ordinal information about individual preferences, so agents' choices reveal the information that is needed to compute these indices.

As is well known, restricting one's attention to incentive-compatible allocations is equivalent to studying income tax schemes. The applications we present

here will turn into identifying properties that optimal tax schemes should satisfy. Here is a preview of the main results of the chapter.

First, we prove that social ordering functions satisfying equal-preferences transfer, the most demanding compensation axiom, call for more redistribution than the social ordering functions satisfying equal-skill transfer, the most demanding responsibility axiom. Second, we prove that all the social ordering functions highlighted here call for nonpositive marginal rates of taxation on low incomes. That is the most important result of the chapter. Both $\mathbf{R}^{s_{\min}^{\text{lex}}}$ and $\mathbf{R}^{\tilde{E}W}$, in spite of the considerable differences in the axioms that justify them, call for a zero marginal rate of taxation, whereas $\mathbf{R}^{s^{\text{lex}}}$ calls for nonpositive marginal tax rates. The third conclusion is that $\mathbf{R}^{\tilde{E}W}$, in spite of its egalitarian flavor, may call for low redistribution.

In Section 11.2, we define the model and study the incentive compatibility constraints. In Section 11.3, we derive the simple criteria that can be used to evaluate current income tax schemes and identify who should benefit in priority from a reform. In Section 11.4, we study some characteristics of the optimal tax schemes in simple economies composed of four types of agents. In Section 11.5, we turn to the characteristics of optimal tax schemes in general economies.

11.2 THE MODEL

We keep the same model as in the previous chapter. An economy is a list $E = (s_N, R_N) \in \mathcal{E}$ where for all $i \in N$, $s_i \in S = [s_{\min}, s_{\max}]$ denotes i 's production skill and R_i i 's preferences over labor time-consumption bundles $z_i = (\ell_i, c_i) \in X = [0, 1] \times \mathbb{R}_+$. There is no constraint on the value of $s_{\min}$, but the case $s_{\min} > 0$ is the most interesting. We come back to that issue in the conclusion of the chapter.

The policymaker has social preferences represented by a SOF $\mathbf{R}$ defined over all allocations. If information were complete, the policymaker could simply maximize $\mathbf{R}$ over the set of feasible allocations. We assume in this chapter that information is not complete. Skills and preferences are private information of the agents. The policymaker knows the different elements of s_N and R_N , but does not know who is who. To state it differently, the policymaker knows only the statistical distribution of types in the economy. Consequently, he or she must restrict attention to incentive-compatible allocations. The contents of the set of incentive-compatible allocations depend on what the policymaker is able to observe. We consider two different informational contexts in turn.

In the first context, the policymaker observes labor time and pretax income for each agent. Given that the policymaker does not observe the agent's skill, the agent can choose to work at a wage rate that is lower than his or her skill, if it is in the agent's interest. We denote by $w_i \leq s_i$ the wage rate at which agent $i \in N$ actually works; that is, w_i corresponds to the gross earnings/labor time ratio of agent i . By convention, we assume that the actual wage rate of an agent who does not work is $s_{\min}$.

An allocation $z_N \in X^N$ is *incentive compatible* if and only if there exists a list of individual wage rates $w_N \in S^N$ at which agents work such that no agent envies the bundle of any other agent working at a wage rate she could earn: for all $i, j \in N$,

$$s_i \geq w_j \Rightarrow (\ell_i, c_i) R_i(\ell_j, c_j). \quad (11.1)$$

We denote by $\widehat{Z}(E)$ the set of feasible incentive-compatible allocations for economy E when the policymaker observes labor times and earnings. Allocations that are Pareto undominated in $\widehat{Z}(E)$ are called *second-best efficient* for E when labor is observable.

There is a very convenient way of describing $\widehat{Z}(E)$. As the policymaker observes ℓ_i and $w_i \ell_i$, he or she can infer w_i . As a result, the policymaker can offer a tax function on labor time, $\tau_w : [0, 1] \rightarrow \mathbb{R}$, that is specific to each value of w . Agent $i \in N$ will then choose w_i and (ℓ_i, c_i) , maximizing the agent's satisfaction subject to the constraint that

$$\begin{aligned} w_i &\leq s_i \\ c_i &\leq w_i \ell_i - \tau_{w_i}(\ell_i). \end{aligned}$$

When $c_i = w_i \ell_i - \tau_{w_i}(\ell_i)$ for all $i \in N$, the allocation is feasible if and only if

$$\sum_{i \in N} \tau_{w_i}(\ell_i) \geq 0.$$

If the inequality is strict, we say that the allocation generates a budget surplus.

Every incentive-compatible allocation can be obtained by such a menu of tax functions. Moreover, as stated in the Lemma 11.1, every incentive-compatible allocation can be obtained in such a way that every $i \in N$ works at $w_i = s_i$ and is therefore submitted to the tax function τ_{s_i} . Intuitively, this is explained as follows. If an allocation is incentive-compatible, there exists, for each class of agents with a given skill s , an opportunity set in the (ℓ, c) space that contains the best bundle of all agents having an equal skill or a lower skill than theirs and that lies everywhere below their indifference curves.¹ The upper frontier of such sets can be used to construct suitable tax functions τ_s .

Lemma 11.1 *Let $E = (s_N, R_N) \in \mathcal{E}$. A feasible allocation $z_N = (\ell_N, c_N)$ for E is incentive-compatible when the policymaker observes labor time and earnings – that is, satisfies (11.1) – if and only if for all $w \in S$ there exists a tax function $\tau_w : [0, w] \rightarrow \mathbb{R}$ such that, for all $i \in N$:*

- (a) $c_i = \ell_i s_i - \tau_{s_i}(\ell_i)$,
- (b) for all $w \leq s_i, \ell \in [0, 1] : (\ell_i, c_i) R_i(\ell, \ell w - \tau_w(\ell))$.

Proof. Only if: Let $E = (s_N, R_N) \in \mathcal{E}$. Let $z_N = (\ell_N, c_N) \in \widehat{Z}(E)$. We need to construct a menu of τ_w satisfying the conditions of the lemma. Let $N_w = \{i \in$

¹ This equivalence between the no-envy property and the existence of a common opportunity set has been mentioned in the early literature on fairness.

$N \mid s_i \geq w\}$. For $w \in S$, let τ_w be such that the graph of $f_w(\ell) = w\ell - \tau_w(\ell)$ in the (ℓ, c) space is the lower envelope of the indifference curves of all individuals $j \in N$ such that $s_j \geq w$; that is,

$$\tau_w(\ell) = \sup\{t \in \mathbb{R} \mid \exists i \in N_w, (\ell, w\ell - t) R_i(\ell_i, c_i)\}.$$

To prove condition (a), we need to prove that for all $i \in N$, $\tau_{s_i}(\ell_i) = s_i\ell_i - c_i$. When $t = s_i\ell_i - c_i$, one has $(\ell_i, s_i\ell_i - t) = (\ell_i, c_i)$, implying $(\ell_i, s_i\ell_i - t) R_i(\ell_i, c_i)$ and therefore

$$s_i\ell_i - c_i \in \{t \in \mathbb{R} \mid \exists j \in N_{s_i}, (\ell_i, s_i\ell_i - t) R_j(\ell_j, c_j)\},$$

so necessarily, $\tau_{s_i}(\ell_i) \geq s_i\ell_i - c_i$. Suppose that $\tau_{s_i}(\ell_i) > s_i\ell_i - c_i$. By definition, there is $j \in N_{s_i}$ such that $(\ell_i, s_i\ell_i - \tau_{s_i}(\ell_i)) R_j(\ell_j, c_j)$ and as $c_i > s_i\ell_i - \tau_{s_i}(\ell_i)$, by monotonicity one has $(\ell_i, c_i) P_j(\ell_j, c_j)$. This is precisely what incentive compatibility excludes.

We prove condition (b) in two steps. First, by construction, no part of the graph of f_{s_i} lies in the strict upper contour set of R_i at (ℓ_i, c_i) , so (ℓ_i, c_i) is among the best bundles for i if i chooses to work at s_i . Second, we need to show that it is in i 's interest to work at s_i . For all $w < w' \in S$, $N_w \supseteq N_{w'}$, so for all $\ell \in [0, 1]$, $\tau_w(\ell) \geq \tau_{w'}(\ell)$ (because $\tau_w(\ell)$ is the supremum of the union of more sets than $\tau_{w'}(\ell)$). Choosing a lower w than s_i only decreases (with respect to inclusion) the budget set of labor time-consumption bundles from which to choose.

If: Let $E = (s_N, R_N) \in \mathcal{E}$. Let $z_N = (\ell_N, c_N)$ be obtained from a menu of tax functions τ_w satisfying the two conditions of the lemma. Let $i, j \in N$ be such that $s_i \leq s_j$. By condition (a), $c_i = s_i\ell_i - \tau_{s_i}(\ell_i)$. By condition (b), $(\ell_j, c_j) R_j(\ell_i, s_i\ell_i - \tau_{s_i}(\ell_i))$, the desired outcome. ■

The consequence of this lemma is that we can restrict our attention to allocations such that each agent i chooses to work at his or her true skill level and freely chooses labor time given how her pretax income will be taxed by function τ_{s_i} .

Let us now model the second informational context that is considered in this chapter. The policymaker observes only earnings – that is, the pretax incomes $w_i\ell_i$. Note that, in this context, it is always best for agent $i \in N$ to earn any given gross income by working at his or her maximal wage rate $w_i = s_i$, as that minimizes the agent's labor time for a fixed level of consumption. We can, therefore, focus on allocations z_N and simply assume that $w_i = s_i$ for all $i \in N$. Also observe that by working $s_j\ell_j/s_i$, agent i obtains the same earnings as agent j .

An allocation $z_N \in X^N$ is *incentive compatible* if and only if no agent envies the earnings-consumption bundle of any other agent whose level of earnings this agent could reach with his or her own skill : for all $i, j \in N$,

$$s_i \geq s_j\ell_j \Rightarrow (\ell_i, c_i) R_i \left(\frac{s_j}{s_i}\ell_j, c_j \right). \quad (11.2)$$

We denote by $\tilde{Z}(E)$ the set of feasible incentive-compatible allocations for economy E when the policymaker observes only earnings. Allocations that are Pareto undominated in $\tilde{Z}(E)$ are called second-best efficient for E when labor is unobservable.

It will prove useful to concentrate on bundles described by earnings rather than by labor time. Let $i \in N$. Let $\ell, \ell' \in [0, 1]$, $y, y' \in \mathbb{R}_+$ be such that $y = s_i \ell$ and $y' = s_i \ell'$. As s_i is fixed, we can slightly abuse notation and write $(y, c) R_i (y', c')$ to denote $(\ell, c) R_i (\ell', c')$.

Again, there is a very convenient way of describing $\tilde{Z}(E)$. As the policymaker observes $y_i = s_i \ell_i$, he or she can propose a tax function on earnings $\tau : \mathbb{R}_+ \rightarrow \mathbb{R}$. Agent $i \in N$ will then choose (y_i, c_i) maximizing his or her satisfaction subject to the constraint

$$c_i \leq y_i - \tau(y_i).$$

When $c_i = y_i - \tau(y_i)$ for all $i \in N$, the allocation is feasible if and only if

$$\sum_{i \in N} \tau(y_i) \geq 0.$$

Again, if the inequality is strict, the tax function is said to generate a budget surplus.

We now claim that any incentive-compatible allocation can be generated by such a tax function. The intuition for this result is identical to that of the previous one. Again, the fact that no-envy is equivalent to the free choice of agents in an equal opportunity set is essential.

Lemma 11.2 *Let $E = (s_N, R_N) \in \mathcal{E}$. A feasible allocation $z_N = (\ell_N, c_N)$ for E is incentive compatible when the policymaker observes only earnings – that is, satisfies (11.2) – if and only if there exists a tax function $\tau : \mathbb{R}_+ \rightarrow \mathbb{R}$ such that, letting $y_i = s_i \ell_i$, for all $i \in N$:*

- (a) $c_i = y_i - \tau(y_i)$,
- (b) for all $y \in [0, s_i]$, $(y_i, c_i) R_i (y, y - \tau(y))$.

The proof is similar to the proof of the previous lemma and is therefore omitted.

In this chapter, we study the properties of the choice of a policymaker when he or she maximizes one of the SOFs we axiomatized in the previous chapter, and when the policymaker faces one of the informational contexts we just described. For the sake of simplicity, given that skill is always assumed to be unobservable and earnings to be observable, we characterize the two informational structures by specifying only whether labor time is observable or not.

As incentive-compatible allocations can all be described in terms of tax functions, the second-best optimality exercise we carry on here is equivalent to evaluating tax functions. This justifies the title of the chapter.

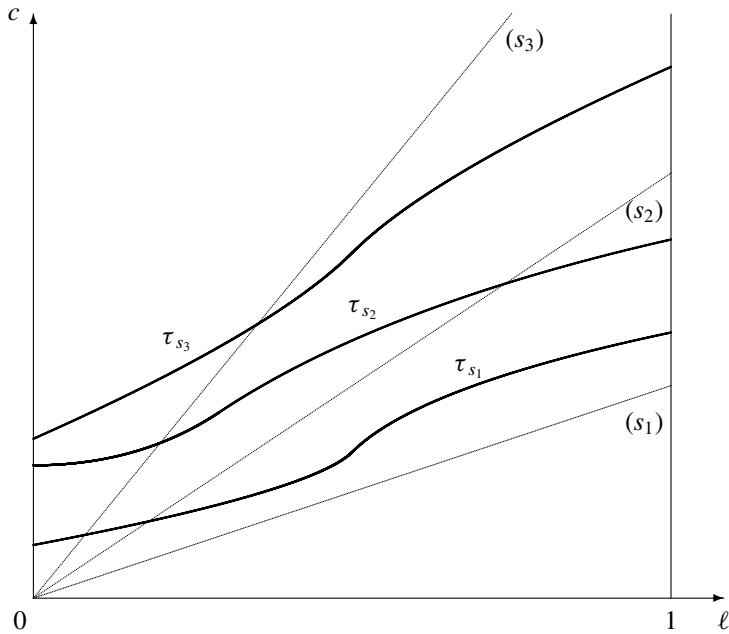


Figure 11.1. A menu of tax functions $\{\tau_s\}$ when labor time is observable

11.3 ASSESSING REFORMS

Our first second-best application deals with the evaluation of reforms. More precisely, let us assume that there is a status quo taxation scheme, and moreover, let us assume that it is not optimal for the SOF the policymaker is interested in. If the status quo must be slightly changed, where should it be changed – that is, who should pay slightly more tax, and who should pay less? We study this question here, in each of the two informational contexts identified in the previous section.

This application will again highlight the advantage of using SOFs built on the leximin aggregator. Indeed, identifying who should benefit from the reform is identical to identifying who is the worst off according to the particular well-being index embodied in the SOF.

Let us start with the case of observable labor time. Following Lemma 11.1, we assume that there exists a menu of tax functions satisfying the conditions listed previously. Figure 11.1 illustrates such a menu in a three-skill economy, with $s_1 < s_2 < s_3$. Thin lines represent the opportunity sets agents would face in the absence of any taxation. The τ_s functions are represented through the consumption functions $s_i \ell_i - \tau_{s_i}(\ell_i)$. There is a discontinuity at $\ell = 0$, as all curves must coincide at that point.

We assume that all points along the τ_s functions are relevant, that is, each point of these curves hits the indifference curve of at least one

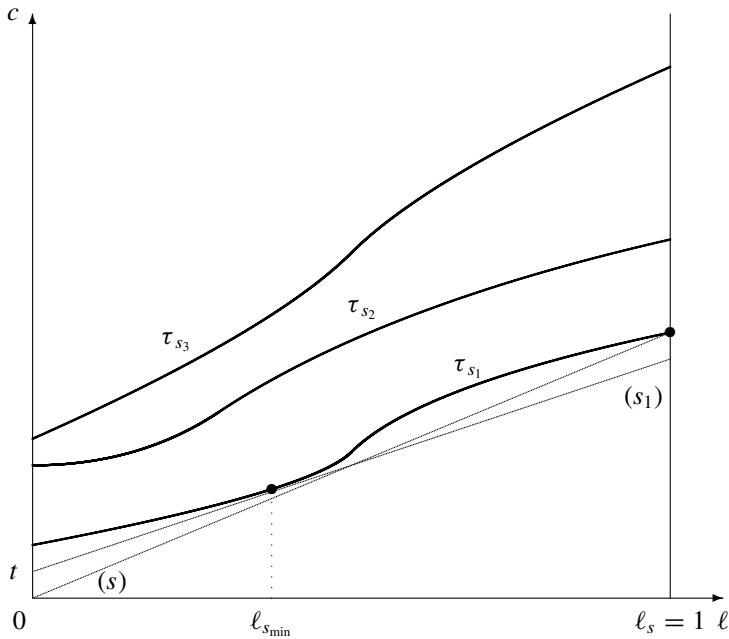


Figure 11.2. Evaluation of tax functions $\{\tau_s\}$ according to $\mathbf{R}^{slex}$ or $\mathbf{R}^{sminlex}$

agent.² Comparing the curves, we can see that all the low-skilled agents are subsidized (they end up with a higher consumption than without taxation), whereas middle- and high-skilled agents are subsidized only if their labor time is low.

Figure 11.2 illustrates how to identify the worst-off agents when the policy-maker's preferences correspond to $\mathbf{R}^{slex}$ or $\mathbf{R}^{sminlex}$. Let us consider $\mathbf{R}^{slex}$ first. The ray of slope s represented in the figure is the highest such ray that is below the graph of all curves. It is tangent to the consumption function associated to τ_{s_1} at $\ell_s = 1$. One can deduce that the indifference curve of agents choosing a labor time of ℓ_s is tangent to the ray of slope s . Their skill-equivalent utility is, therefore, equal to s . By a simple revealed-preferences argument, the graph shows that all other agents have a skill-equivalent utility at least as large as s . That proves that, for $\mathbf{R}^{slex}$, low-skilled agents working full time are, in this example, the ones who should benefit from a reform. It need not be the case, in general, that hard-working low-skilled agents are the worst-off agents according to $\mathbf{R}^{slex}$. The key parameter is the average consumption per unit of labor, $(s_i \ell_i - \tau_{s_i}(\ell_i)) / \ell_i$.

² In the case of a finite set of agents, this is achieved by assuming that the graph of the consumption function associated with τ_s functions is the lower envelope of the indifference curves of all agents with that skill.

Let us now turn to $\mathbf{R}^{s_{\min}^{\text{lex}}}$. In Figure 11.2, we claim that the worst-off agents according to $\mathbf{R}^{s_{\min}^{\text{lex}}}$ are the low-skilled agents choosing to work $\ell_{s_{\min}}$. Indeed, by a similar argument as previously, the indifference curve of agents choosing that labor time is tangent to the budget of slope $s_{\min}$ corresponding to a lump-sum subsidy of t . Moreover, the implicit budget of slope $s_{\min}$ of any agent choosing a different labor time is above the budget of that slope represented in the figure. That proves the claim. More generally, the worst-off agent is the low-skilled agent receiving the lowest subsidy, or equivalently, “paying” the largest tax, $\tau_{s_{\min}}(\ell)$.

Let us study the generality of two properties of the preceding example. First, both $\mathbf{R}^{s^{\text{lex}}}$ and $\mathbf{R}^{s_{\min}^{\text{lex}}}$ force us to conclude that worst-off agents are those with minimal skill. This property is, actually, fully general. The incentive-compatibility constraints, indeed, protect higher-skilled agents and provide them with larger opportunity sets, as proved in Lemma 11.1. It is therefore impossible for a higher-skilled agent to end up with a lower s -equivalent or $s_{\min}$ -equivalent well-being level than the worst-off among the low-skilled agents.

Second, the labor time of the worst-off agents according to $\mathbf{R}^{s^{\text{lex}}}$ is larger than that according to $\mathbf{R}^{s_{\min}^{\text{lex}}}$. This comes from the fact that t , in the graph, is positive, so the slope of the ray that is used to identify the worst-off according to $\mathbf{R}^{s^{\text{lex}}}$ is steeper than $s_{\min}$. It is easy to figure out that in the case $t = 0$, the two lines would coincide, so the worst-off agents would be the same. From this observation, we can conclude that if $\tau_{s_{\min}}$ is such that all the low-skilled agents are subsidized, then a policymaker maximizing $\mathbf{R}^{s^{\text{lex}}}$ will focus on low-skilled agents working at least as much as those on which a policymaker maximizing $\mathbf{R}^{s_{\min}^{\text{lex}}}$ would focus. If $\tau_{s_{\min}}$ is such that some low-skilled agents pay a strictly positive tax, then the conclusion is reversed.

Figure 11.3 illustrates how to identify the worst-off agents according to $\mathbf{R}^{\tilde{\ell}^{EW}}$ for two values of $\tilde{\ell}$, $\tilde{\ell}^* < \tilde{\ell}^{**}$. First, identifying which agents, in each skill subgroup, is worse off, does not depend on $\tilde{\ell}$. Among low-skilled agents, the implicit budget of the worst-off is the same as when we apply $\mathbf{R}^{s_{\min}^{\text{lex}}}$. In this example, they are the agents choosing a labor time of $\ell_{s_{\min}} = \tilde{\ell}^*$. Among higher-skilled agents, however, the implicit budgets are different, as $\mathbf{R}^{\tilde{\ell}^{EW}}$ recommends to use the agents' actual skills to evaluate their well-being. In the graphical example, among s_2 and s_3 agents, the worst-off agents are those working full time.

Second, we have to compare the well-being of the worst-off across skill subgroups. The consequence of using implicit budgets of different slopes to evaluate the well-being of agents of different skills is drastic. With a low value of $\tilde{\ell}$, such as $\tilde{\ell}^*$, the worst-off agents in the economy are the high-skilled agents working full time. Their $\tilde{\ell}$ -equivalent well-being index is

$$\begin{aligned} c_i - s_3 \ell_i + s_3 \tilde{\ell}^* &= s_3 \ell_i - \tau_{s_3}(\ell_i) - s_3 \ell_i + s_3 \tilde{\ell}^* = s_3 \tilde{\ell}^* - \tau_{s_3}(\ell_i) \\ &= s_3 \tilde{\ell}^* - \tau_{s_3}(1). \end{aligned}$$

The $\mathbf{R}^{\tilde{\ell}^{EW}}$ SOF recommends in this example that the reform benefit the agents with the largest income. With a higher value of $\tilde{\ell}$, such as $\tilde{\ell}^{**}$, the ranking of

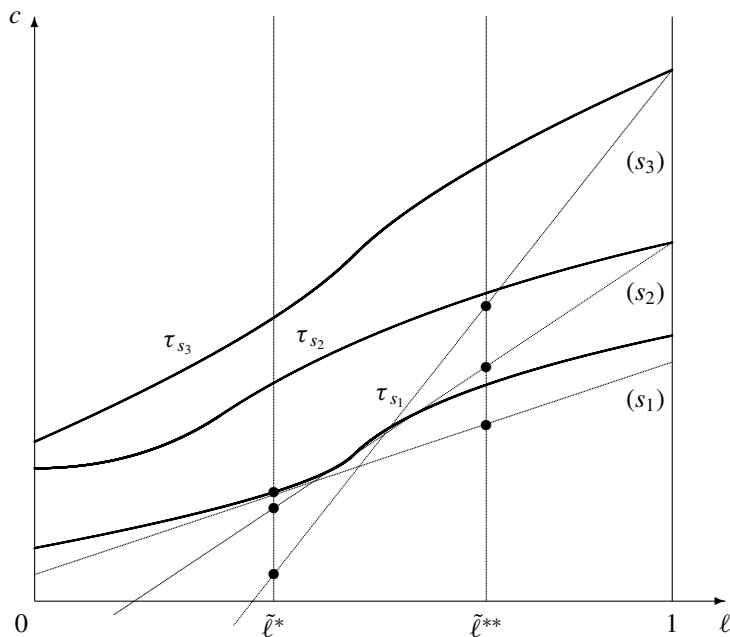


Figure 11.3. Evaluation of tax functions $\{\tau_s\}$ according to $\mathbf{R}^{\tilde{\ell}EW}$

the worst-off agents from each subgroup is different: low-skilled agents turn out to be worst-off. Their $\tilde{\ell}$ -equivalent well-being index is now

$$\begin{aligned} c_i - s_1 \ell_i + s_1 \tilde{\ell}^{**} &= s_1 \ell_i - \tau_{s_1}(\ell_i) - s_1 \ell_i + s_1 \tilde{\ell}^{**} = s_1 \tilde{\ell}^{**} - \tau_{s_1}(\ell_i) \\ &= s_1 \tilde{\ell}^{**} - \tau_{s_1}(\ell_{s_{\min}}). \end{aligned}$$

The reform should then benefit low-skilled agents choosing a labor time of $\ell_{s_{\min}}$.

Three general features of reforms consistent with $\mathbf{R}^{\tilde{\ell}EW}$ can be derived from this example. First, it need no longer be the case that the worst-off agents in the economy are to be found among the minimal-skilled agents. It may even be the case that they are to be found among the highest-skilled agents. Second, an increase in the reference value $\tilde{\ell}$ benefits the low-skilled agents: the larger $\tilde{\ell}$, the lower the skill of the subgroup of agents on which the reform should focus. Third, the simple computation to make to identify the agents who should benefit in priority from the reform is to look for the lowest $s\tilde{\ell} - \tau_s(\ell)$ among all s and among all ℓ .

Let us now turn to the assumption of nonobservable labor time. Following Lemma 11.2, we assume that there is an income tax function τ , and we imagine that it represents the status quo from which a reform could depart. We need to identify the worst-off agents according to each of our SOFs.

Figure 11.4 illustrates a tax function in a similar three-skill economy as previously. The consumption function that is depicted has equation $y - \tau(y)$, so

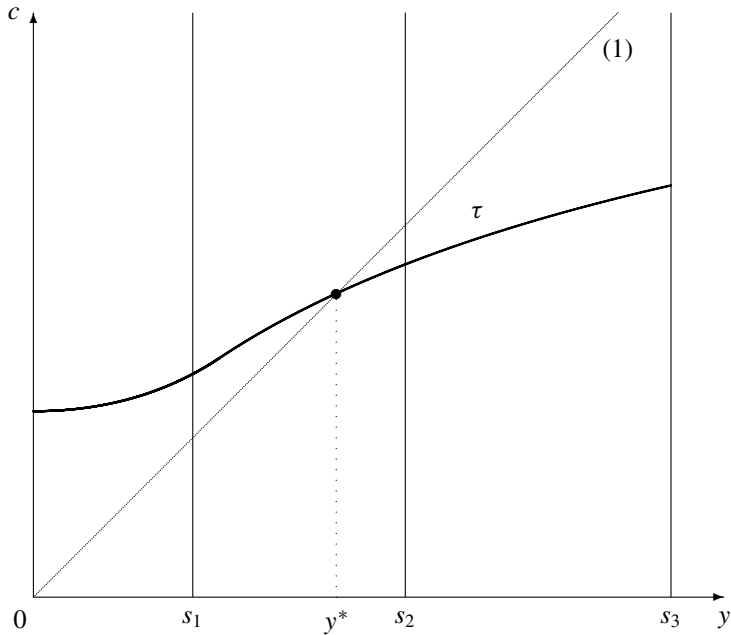


Figure 11.4. A tax function τ_s when labor time is unobservable

the tax corresponding to a level of pretax income is the difference between the line of slope 1 (that is, the no-taxation line) and the consumption level associated with that particular pretax income. Because of the incentive-compatibility constraints embedded in the figure, there is no loss of generality in assuming that this consumption function $f(y) = y - \tau(y)$ is nondecreasing. An earnings-consumption bundle in a strictly decreasing segment, indeed, could never be chosen by a rational agent, as this agent could get a larger consumption with a lower labor time.

In this example, all low-skilled agents receive a subsidy (they pay a negative tax). Among middle-skilled agents (respectively, high-skilled agents), those having a pretax income lower than y^* , that is, those working less than y^*/s_2 (respectively, y^*/s_3), receive a subsidy, and the others pay a tax. As earlier, we assume that all the points along the curve are relevant; that is, each point of the curve hits the indifference curve of at least one agent. Moreover, we assume that for each s_i , each point of the curve for incomes lower than s_i hits the indifference curve of at least one agent having this skill.

Figure 11.5 illustrates how to identify the worst-off agents when the policy-maker's objectives correspond with $\mathbf{R}^{\text{lex}}$. We need to distinguish two steps. First, the worst-off agents need to be identified in each skill subgroup. This is done in the same way as under the assumption of an observable labor time. In this example (and for the sake of clarity), it turns out that the worst-off among agents of each subgroup are the ones working full time. Second, we

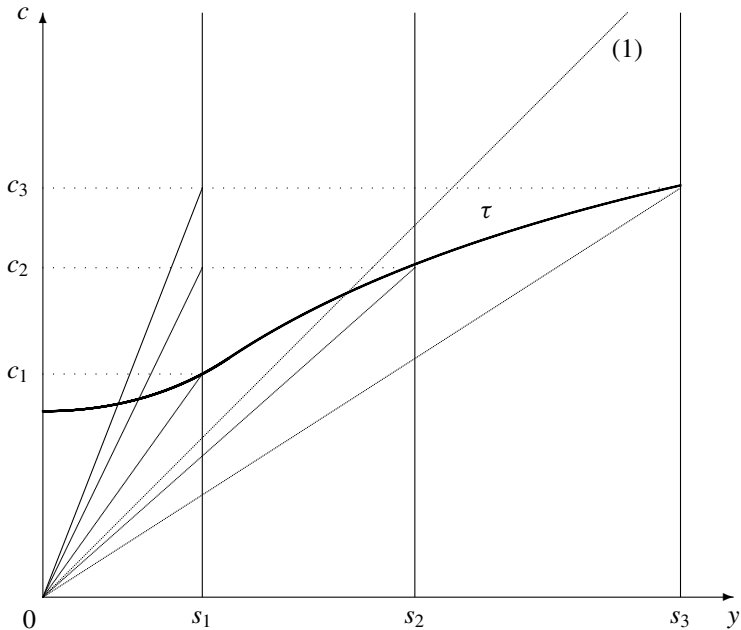


Figure 11.5. Evaluation of tax function τ according to $\mathbf{R}^{slex}$

need to compare the s -equivalent well-being levels among the worst-off of each subgroup. In the graphical example, for each subgroup $j \in \{1, 2, 3\}$, that level is given by c_j/s_j . These numbers are not immediately comparable, as the denominators are expressed in terms of pretax income instead of labor time.

There is an easy way of comparing them, which involves rescaling them to express them in terms of a common labor time unit. This is illustrated in Figure 11.5. The ray of slope c_j/s_j for middle and high-skilled agents is rescaled so as to fit the way labor time is measured for low-skilled agents. We are left with comparing the quantities c_1/s_1 , $c_2/(\frac{s_1}{s_2}s_2) = c_2/s_1$, and $c_3/(\frac{s_1}{s_3}s_3) = c_3/s_1$. Consequently, the simple observation that $c_1 \leq c_2 \leq c_3$ leads us to conclude that the worst-off agents among the low-skilled ones are also the worst-off in the entire population. Again, higher-skilled agents are protected by the incentive-compatibility constraints and can never end up enjoying a lower well-being level than the worst-off among the low-skilled agents.

In this example, therefore, the policymaker should reform the tax system so as to decrease the tax (increase the subsidy) of the low-skilled agents working full time. More generally, those who should benefit from a tax reform can be found by minimizing $(y - \tau(y))/y$, or simply maximizing $\tau(y)/y$, among the low-skilled agents – that is, among the pretax income levels $y \leq s_{\min}$.

Figure 11.6 illustrates how to identify the worst-off agents when the policymaker's preferences correspond to $\mathbf{R}^{s_{\min}lex}$. It is again convenient to distinguish two steps. First, the worst-off agents need to be identified in each skill subgroup.

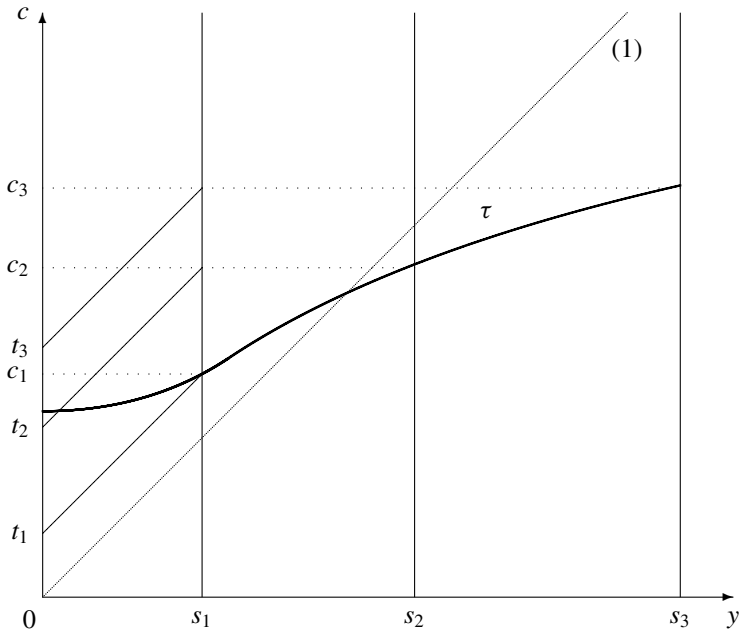


Figure 11.6. Evaluation of tax function τ according to $\mathbf{R}^{s_{\min}\text{lex}}$

This is done in the same way as under the assumption of an observable labor time, and as in the case of $\mathbf{R}^{s\text{lex}}$ in the previous paragraphs, the worst-off among agents of each subgroup, in this example, are the ones working full time. Second, we need to compare the $s_{\min}$ -equivalent well-being levels among the worst-off agents from each subgroup. In each subgroup $j \in \{1, 2, 3\}$, that level is given by $c_j - s_{\min}\ell_j = c_j - s_{\min}$, as $\ell_j = 1$ for the worst-off agents. These operations are illustrated in the figure in the following way: Earnings-consumption bundles are first rescaled so they fit the low-skill consumption set. Then the three parallel lines correspond to the subtraction of $s_1 = s_{\min}$, and we get t_1 , t_2 , and t_3 .

Again, nondecreasingness of the consumption function guarantees that $c_1 \leq c_2 \leq c_3$, which, in turn, is necessary and sufficient to get $t_1 \leq t_2 \leq t_3$. Again, the worst-off in the population must be found among low-skilled agents. In this example, therefore, the policymaker should reform the tax system to decrease the tax (increase the subsidy) of the low-skilled agents working full time. More generally, the beneficiary of a desirable tax reform can be found by minimizing $c - s_{\min}\ell = y - \tau(y) - s_{\min}(y/s_{\min}) = -\tau(y)$, or simply maximizing $\tau(y)$, among pretax income levels $y \leq s_{\min}$.

In our example, both $\mathbf{R}^{s\text{lex}}$ and $\mathbf{R}^{s_{\min}\text{lex}}$ force us to look at the shape of the tax function on low incomes. This is a fully general property, directly driven from the combination of the incentive-compatibility constraints and the fact that the well-being index we use does not depend on the agents' actual skills.

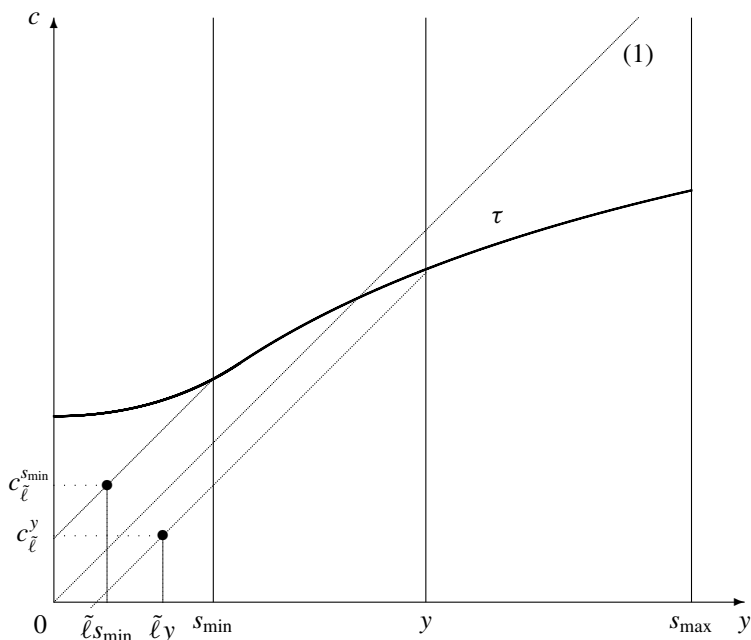


Figure 11.7. Evaluation of tax function τ according to $\mathbf{R}^{\tilde{E}W}$.

The two SOFs also force us, in the examples given here, to look at low-skilled agents working full time. This, however, is not general. It may even be the case that $\mathbf{R}^{s_{\min}^{\text{lex}}}$ and $\mathbf{R}^{s_{\min}^{\text{lex}}}$ force us to look at agents with different labor time. Again, the key condition is whether all low-skilled agents receive a subsidy. In this case, $\mathbf{R}^{s_{\min}^{\text{lex}}}$ focuses on agents working at least as much as those on which $\mathbf{R}^{s_{\min}^{\text{lex}}}$ focuses. If some low-skilled agents pay a positive tax, then the opposite relation holds.

Let us now turn to $\mathbf{R}^{\tilde{E}W}$ and show how it can be used to assess reforms. Figure 11.7 illustrates how to identify the worst-off agents when the policy-maker's preferences correspond to $\mathbf{R}^{\tilde{E}W}$.

Again, in a first step, the worst-off agents need to be identified in each skill subgroup. Among the minimal-skilled agents, we need to construct implicit budgets that have a slope of $s_{\min}$. This is done in the same way as for $\mathbf{R}^{s_{\min}^{\text{lex}}}$. In our example, the agents working full time (or being indifferent between their current bundle and working full time) have the lowest implicit budget. Should an agent, facing this budget, choose a labor time of $\tilde{\ell}$, she would enjoy a consumption level of $c_{\tilde{\ell}}^{s_{\min}} = s_{\min}\tilde{\ell} - \tau(s_{\min})$. This quantity measures the $\tilde{\ell}$ -equivalent well-being index of the worst-off agents in the subgroup of minimal-skilled agents.

For any $y \in [s_{\min}, s_{\max}]$ such that there are agents with a skill level equal to y , we can compute the corresponding well-being level in a similar way,

Table 11.1. *Criteria to Evaluate Reforms, as a Function of the SOF and the Information Constraints*

	Observable labor	Unobservable labor
$\mathbf{R}^{s_{\min} \text{lex}}$	$\min_{\ell \in [0,1]} -\tau_{s_{\min}}(\ell)$	$\min_{y \in [0,s_{\min}]} -\tau(y)$
$\mathbf{R}^{s \text{lex}}$	$\min_{\ell \in [0,1]} \frac{-\tau_{s_{\min}}(\ell)}{\ell}$	$\min_{y \in [0,s_{\min}]} \frac{-\tau(y)}{y}$
$\mathbf{R}^{\tilde{\ell}EW}$	$\min_{s \in S, \ell \in [0,1]} (\tilde{\ell}s - \tau_s(\ell))$	$\min_{s \in S, y \in [0,s]} (\tilde{\ell}s - \tau(y))$

as exemplified in the figure. Among agents having a skill level equal to y , those working full time (or being indifferent between their current bundle and working full time) are the worst off. When evaluated according to $\mathbf{R}^{\tilde{\ell}EW}$, the implicit budget of agents of skill y working full time has a slope of y . That is the key difference from the previous SOFs. In the (y, c) -space, such a budget, represented from a point of y coordinate, has a slope of 1. Should an agent, facing this budget, choose a labor time of $\tilde{\ell}$, the agent would enjoy a consumption level of $c_{\tilde{\ell}}^y = \tilde{\ell}y - \tau(y)$. This quantity measures the $\tilde{\ell}$ -equivalent well-being index of the worst-off agent in the subgroup of y -skilled agents.

Remember that it need not be the case that worst-off agents are those (indifferent to) working full time. In our example, it follows from the fact that the slope of the consumption function is lower than 1, which corresponds to a positive marginal tax rate. This is clearly the most relevant case, but it is not the only one. The most general result is that, among agents having a skill level equal to s , the worst-off have a well-being index equal to $\min_{y \in [0,s]} (\tilde{\ell}s - \tau(y))$. It is now easy to define the criterion that needs to be used to find the worst-off among the whole population: agents $i \in N$ having the lowest $\min_{y \in [0,s_i]} (\tilde{\ell}s_i - \tau(y))$ need to benefit in priority from a reform.

As it is clear from Figure 11.7, higher-skilled agents may have a lower well-being index than minimal-skilled agents: $c_{\tilde{\ell}}^y < c_{\tilde{\ell}}^{s_{\min}}$. This conclusion is similar to the one we reached under the assumption that labor time is observable: whereas our two SOFs grounded on the axiom of equal-preferences transfer (i.e., $\mathbf{R}^{s \text{lex}}$ and $\mathbf{R}^{s_{\min} \text{lex}}$) force us to find the worst-off agents among the low-skilled, the $\mathbf{R}^{\tilde{\ell}EW}$ family of SOFs, grounded on the axiom of equal-skill transfer, may lead us to claim that some higher-skilled agents pay too much income tax. Let us, again, observe that the latter conclusion depends on the value of $\tilde{\ell}$. In the example, $\tilde{\ell}$ is rather low (around 1/3). For larger values of $\tilde{\ell}$, the well-being index values will change, and the ranking may change. For sufficiently large values of $\tilde{\ell}$, the low-skilled agents will be the worst off.

Let us now summarize the results obtained in this section about the evaluation of reforms using the SOFs characterized in the previous chapter.

- As a function of the SOF one is interested in, and as a function of the informational structure one faces, the worst-off agents are identified by minimizing the criteria in Table 11.1.
- Independent of the informational structure, both $\mathbf{R}^{s \text{lex}}$ and $\mathbf{R}^{s_{\min} \text{lex}}$ – that is, our SOFs satisfying equal-preferences transfer – recommend

focusing on low-skilled agents. On the contrary, $\mathbf{R}^{\tilde{E}W}$, satisfying equal-skill transfer, may recommend that the reform benefit higher-skilled agents.

- Among low-skilled agents, $\mathbf{R}^{slex}$ recommends that the reform benefit agents who work at least as much as those who should benefit from a reform based on $\mathbf{R}^{sminlex}$ if all low-skilled agents are currently subsidized (and, conversely, in the opposite case).

This section has illustrated that our SOFs can be used to derive simple criteria ready to be used to evaluate reforms. The exercise consists of identifying the agents who should benefit, in priority, from the reform. This exercise is made particularly easy by the fact that our SOFs are of the leximin type. Those who should benefit from a reform are those currently experiencing the lowest well-being index. Because of the constraints imposed by incentive compatibility, they are identified easily by analyzing the shape of the tax functions, which is a striking fact for the practical implementation of such criteria. No information about the distribution of characteristics of the population is needed; knowing the tax rules is enough.

11.4 OPTIMAL TAXATION SCHEMES IN A SIMPLE ECONOMY

The classical literature on optimal taxation has focused mostly on the design of the optimal tax scheme. A taxation scheme is optimal when it maximizes social preferences under incentive compatibility constraints. In this section and the next one, we derive some properties of the optimal tax schemes when social preferences are captured by the SOFs we characterized in Chapter 10 – $\mathbf{R}^{slex}$, $\mathbf{R}^{sminlex}$, and $\mathbf{R}^{\tilde{E}W}$. In this section, we begin our exploration with simple economies composed of four types of agents, similar to the economies we used in the previous chapter to illustrate the three SOFs in which we are interested. The study of a more general model will be tackled in the next section.

There are two possible skills, $s_1 < s_2$, and two possible preferences in the (ℓ, c) space, R_1 and R_2 , such that R_1 exhibit a lower willingness to work than R_2 (that is, facing identical opportunity sets, a R_1 agent would always choose a lower labor time).³

Studying the optimal menu of tax functions when labor time is observable does not bring more insights and is not easier in this simple framework than in the general model. We postpone the study of this case to the next section. We therefore assume that labor time is not observable, so the tax can depend only on earned income, y_i . We have four types of agents; their preferences in the (y, c) space can be denoted R_{11} , R_{12} , R_{21} , and R_{22} , where the first index refers to skill and the second one to preferences. In the graphical illustrations, we assume that the four subgroups have the same size.

³ This corresponds to the Spence–Mirrlees single-crossing condition in the (ℓ, c) space among agents with the same skill.

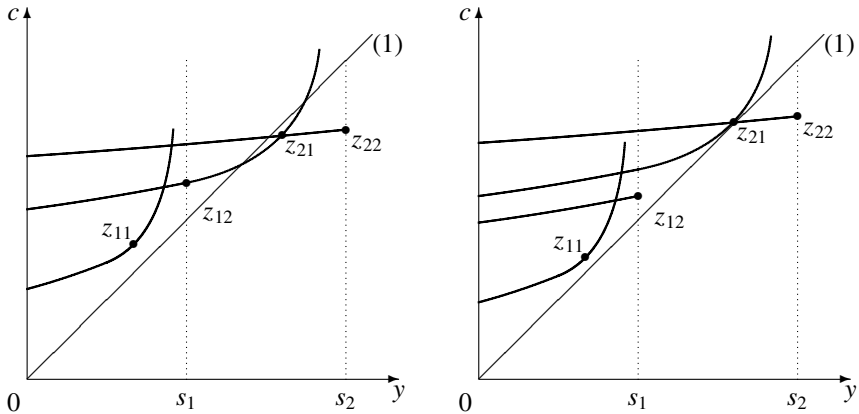


Figure 11.8. Incentive-compatible allocations in a simple economy where $z_{12} \neq z_{21}$

To make the problem of incentive compatibility more interesting, we assume that low-skilled high-willingness-to-work agents cannot be distinguished from high-skilled low-willingness-to-work ones. That is, for any bundles (y, ℓ) , (y', ℓ') that are affordable to both types (that is, such that $y, y' \leq s_1$),

$$(y, \ell) R_{12} (y', \ell') \Leftrightarrow (y, \ell) R_{21} (y', \ell').$$

An allocation in that economy is a list $(z_{ij} = (y_{ij}, c_{ij}))_{i,j \in \{1,2\}}$. Figure 11.8 illustrates two incentive-compatible allocations where $z_{12} \neq z_{21}$. In the left-hand part of the figure, the two bundles are different, but agents of type 21 are indifferent between them. In the right-hand part, they strictly prefer the bundle they are assigned. The possibility of having $z_{12} \neq z_{21}$ comes from our assumption of an upper bound on labor time. In the classical literature, there is no such bound (and, therefore, nothing like a full-time job). To simplify the exposition and to be able to compare our results with the literature, we add the assumption that $z_{12} = z_{21}$.

Optimal allocations for $\mathbf{R}^{\text{lex}}$ and $\mathbf{R}^{\text{minlex}}$ are illustrated in Figure 11.9. In the left-hand part of the figure, the allocation $(z_{11}, z_{12}, z_{21}, z_{22})$, with $z_{12} = z_{21}$, is optimal for $\mathbf{R}^{\text{lex}}$. The argument can be developed geometrically. First, the indifference curves of both types of low-skilled agents are tangent to the ray of slope s . This implies that they have the same well-being index, and it is equal to s . A ray of slope $s \frac{s_1}{s_2}$ is also drawn in the figure. It represents the implicit budget that agents of types 21 and 22 would face, should they enjoy a well-being of s as well. We see in the figure that both types of agents strictly prefer their bundle to choosing in that budget. Their well-being is therefore greater than s . The represented allocation does not succeed in equalizing the well-being indices of all the agents. One would like to further increase the well-being of types 11 and 12, at the expense of the others. However, it is impossible to decrease the well-being of type 21, as it is indistinguishable from type 12 (any deviation would

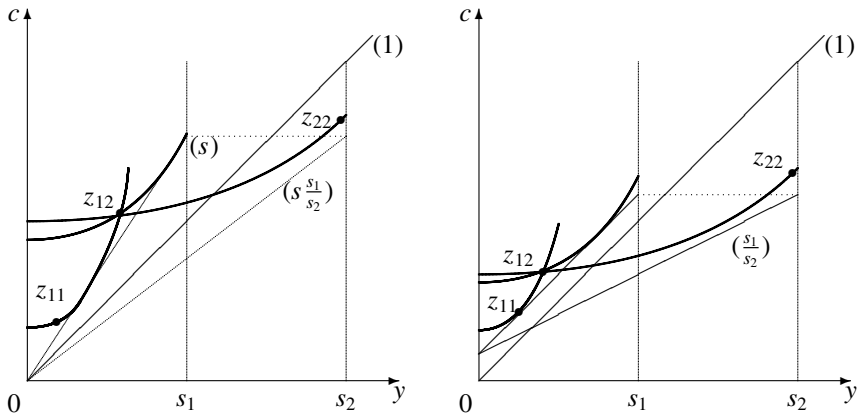


Figure 11.9. Optimal second-best allocations according to $\mathbf{R}^{slex}$ (left) and $\mathbf{R}^{sminlex}$ (right)

violate the incentive compatibility constraint). It is also impossible to decrease the well-being of the agents of type 22. Indeed, as they are indifferent between z_{12} and z_{22} , any decrease in their well-being would give them the incentive to claim that they are of type 12. The incentive-compatibility constraint prevents us from improving the fate of the low-skilled agents. If such an allocation is second-best efficient – in the figure, it is – then it is optimal.

In the right-hand part of Figure 11.9, the allocation $(z_{11}, z_{12}, z_{21}, z_{22})$, with $z_{12} = z_{21}$, is optimal for $\mathbf{R}^{sminlex}$. Again, the indices are equalized among low-skilled agents, which is illustrated by the fact that both indifference curves are tangent to the same line of slope 1, which is equivalent to a slope of $s_{min} = s_1$ in their (ℓ, c) -space. A similar implicit budget for the high-skilled agents is also represented in the figure, with a slope of s_1/s_2 . As in the analysis relative to $\mathbf{R}^{slex}$ in the previous paragraph, we observe that types 21 and 22 are strictly better off than if they were given the opportunity to choose from that budget. Their well-being index is therefore strictly larger than the low-skilled agents' index. We also observe that all incentive-compatibility constraints are binding, which, again, proves that if this allocation is second-best efficient – in the figure, it is – then it is optimal.

In Figure 11.10, the allocation $(z_{11}, z_{12}, z_{21}, z_{22})$, with $z_{12} = z_{21}$, is optimal for $\mathbf{R}^{\tilde{\ell}EW}$, for the represented value of $\tilde{\ell}$. Both low-skilled agents maximize over a budget of slope 1, so, again, their well-being index is equalized. The value of their well-being index is the consumption level they could afford by having a labor time of $\tilde{\ell}$ over that budget – that is, by consuming $\tilde{\ell}s_1$, represented in the figure.

Agents of type 22 are consuming z_{22} , a bundle they strictly prefer to $z_{12} = z_{21}$. The incentive compatibility constraint preventing agents of type 22 to claim they are of type 12 is not binding. That does not mean, though, that the well-being of type 12 should be increased. Indeed, agents of type 22 have exactly

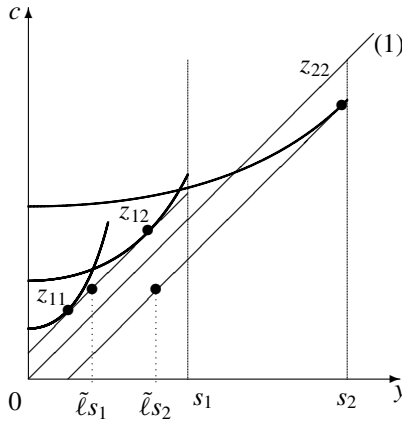


Figure 11.10. Optimal second-best allocations according to $\mathbf{R}^{\tilde{l}EW}$

the same well-being. This is computed by considering the implicit budget to which agents 22 are indifferent, represented in the figure. Given that budget (of slope 1), the consumption level $\tilde{l}s_2$ these agents would obtain by working $\tilde{l}$ is exactly the same as the low-skilled agents' level. Any further redistribution from high-skilled to low-skilled agents would therefore decrease the minimal well-being index in the population. Let us even observe that this allocation is first-best efficient (all slopes of the indifference curves are equal to 1 in the (y, c) -space).⁴ This completes the proof that this allocation is optimal.

A larger value of $\tilde{l}$ would have increased the level of redistribution from agents of type 22 to low-skilled agents, with the consequence that the incentive-compatibility constraint would end up being binding. A lower value of $\tilde{l}$ would have decreased redistribution even more. At the limit, when $\tilde{l} = 0$, the policy of no redistribution – that is, laissez-faire – would be optimal for $\mathbf{R}^{\tilde{l}EW}$. This observation confirms that SOFs based on the leximin criterion do not necessarily call for a large redistribution.

Having identified the properties of optimal second-best allocations for our SOFs, we can deduce the following results:

- The incentive-compatibility constraints of high-skilled agents are always binding when the policymaker maximizes a SOF satisfying equal-preferences transfer. This follows from the fact that absolute priority is given to low-skilled agents. This property does not necessarily hold if the policymaker maximizes a SOF satisfying equal-skill transfer.
- Independently of which SOF is maximized, taxation is never progressive among low-skilled agents. Indeed, the amount transferred to

⁴ With other values of $\tilde{l}$ or in other economies, the optimal allocation for $\mathbf{R}^{\tilde{l}EW}$ need not be first-best efficient.

low-skilled hardworking agents, type 12 agents, is always at least as large as the amount transferred to low-skilled low-willingness-to-work agents, type 11 agents.⁵ Moreover, in the case of $\mathbf{R}^{\text{sllex}}$, it is strictly larger. This result will be confirmed in the next section.

- Finally, even if our social preferences are of the leximin type, they may lead to a very limited redistribution, as exemplified by the $\mathbf{R}^{\tilde{\ell}EW}$ SOF with a small $\tilde{\ell}$. The intuition for this result is the following: The well-being index associated with the $\mathbf{R}^{\tilde{\ell}EW}$ SOF is equalized among agents when all agents working $\tilde{\ell}$ have the same consumption level, independent of their skill. For lower values of $\tilde{\ell}$, the redistribution necessary to equalize that consumption level among agents working $\tilde{\ell}$ is also lower. At the limit, if the policymaker wishes to equalize the consumption of all agents choosing not to work, then no redistribution is necessary.

Boadway et al. (2002) study the same two-skill two-preferences model under the assumptions that preferences are quasilinear in leisure and that there is no upper bound on the labor time. They study utilitarian social preferences; that is, they equip each set of preferences R_1 and R_2 with a numerical representation u_1 and u_2 , linear in consumption, and social welfare is defined as a sum of individual utility levels. They show that when social preferences are utilitarian, the optimal allocations can be associated with a list of binding incentive-compatibility constraints very different from what we obtain here. In particular, it may be the case that redistribution benefits the agents of type 22. This cannot occur here because these agents always have the greatest index of well-being at the laissez-faire allocation, and we apply the maximin criterion. This proves that allocations maximizing one of the social preferences discussed in this chapter form a strict subset of the second-best efficient allocations.

Closer to what we develop here, they study the consequences of a family of utilitarian social preferences that give absolute priority to low-skilled agents in the sense that the weight assigned to the others' utility is zero, whereas the weights given to the utilities of agents with different preferences, among the low-skilled, may vary. Boadway et al. then also reach the conclusion that redistribution may be regressive among low-skilled agents. This illustrates another difference between our approach and classical taxation theory. In the latter, different conclusions are drawn about the optimal tax as a function of the weights that are assigned to different agents. These weights represent the policymaker's ethical preferences. In the former, different conclusions are drawn as a function of which SOFs is maximized – that is, as a function of which axioms capture the policymaker's ethical preferences.

⁵ Recall that the transfers are measured by the vertical distance between the bundles agents are assigned and the no-tax 45° line.

11.5 OPTIMAL TAXATION SCHEMES

In this section we turn to the general model. All results presented here are drawn from Fleurbaey and Maniquet (2006, 2007, 2011). Formal proofs of the results can be found in those sources. We focus here on intuitive presentations.

There are two main differences between the SOFs we study in this section and the typical social preferences that are used in the classical literature on optimal taxation. The first difference comes from the aggregator. All our SOFs are of the leximin (or maximin) type, whereas the literature has concentrated on utilitarian SOFs.⁶ The second difference comes from the indices that are aggregated. Our SOFs aggregate indices of well-being that are axiomatically derived from some notion of equality of resources. The classical utilitarian SOFs aggregate utility levels coming from some exogenous utility function. In the latter case, the most frequent assumption is that all agents have the same utility function (and, therefore, the same preferences). One of the main achievements of the classical literature has been to provide us with the exact formula of the optimal tax scheme under various assumptions.

As it is clear, now, a key feature of our taxation model is that agents differ in both their skill and their preferences. This double heterogeneity has a direct consequence on our objective: it becomes almost impossible to derive the formula of the optimal tax scheme analytically. Optimal tax schemes must be derived from the maximization of the criteria we defined in Section 11.3 under the proper incentive compatibility constraints. As we insisted earlier, these criteria are simple. On the other hand, characterizing the set of incentive-compatible allocations to be able to find the best allocation is an extremely difficult task. This explains why some of our results will be stated in the following format: if this or that tax scheme is second-best efficient, then it is optimal.

On the other hand, the tax schemes we will be able to point out have several properties that can be related to our objective of fairness. Identifying these properties is a key goal of second-best applications of our SOF approach. The main properties we will identify will be a zero or negative marginal tax rate on low incomes.⁷

Let us begin with the case of observable labor time. Our first formal result is that an optimal allocation for $\mathbf{R}^{\tilde{E}W}$ or $\mathbf{R}^{s_{\min}, \text{lex}}$ can be obtained by a menu $\{\tau_s\}$ such that the individuals with the lowest skill face a zero marginal tax.

⁶ There are, however, important studies of the maximin criterion in Atkinson (1973, 1995) and Bowdway and Jacquet (2008), among others.

⁷ In recent developments of optimal taxation theory (Choné and Laroque 2005, Lee and Saez 2008, Kaplow 2008), the assumption of identical preferences has been dropped, and some consequences of the heterogeneity in agents' preferences on the shape of optimal tax schemes have been identified. In the models focusing on participation (the agents choosing only whether they work full time or do not work at all), it has appeared that a negative marginal tax rate may be second-best efficient (whereas the key result of Mirrlees' [1971] seminal paper was that it had to be positive or zero).

With $\mathbf{R}^{\text{slex}}(E)$, the average marginal tax for the low-skilled agents must be nonpositive.

Theorem 11.1 *Assume labor time is observable. Let $E = (s_N, R_N) \in \mathcal{E}$. Every second-best optimal allocation for $\mathbf{R}^{\text{sminlex}}(E)$ or $\mathbf{R}^{\tilde{E}W}(E)$ can be obtained by a menu $\{\tau_s\}$ such that $\tau_{s_{\min}}$ is a nonpositive constant-valued function. Every second-best optimal allocation for $\mathbf{R}^{\text{slex}}(E)$ can be obtained by a menu $\{\tau_s\}$ such that $\tau_{s_{\min}}$ satisfies $\tau_{s_{\min}}(\ell) \geq \tau_{s_{\min}}(1)$ for all $\ell \in [0, 1]$.*

The intuition for this result goes as follows. Let z_N be an optimal allocation for $\mathbf{R}^{\text{sminlex}}(E)$. Following the proof of Lemma 11.1, one can construct the menu $\{\tau_s\}$ that implements this allocation in such a way that higher-skilled agents get better budget sets – that is, for all $\ell \in [0, 1]$, all $s > s'$, $s\ell - \tau_s(\ell) \geq s'\ell - \tau_{s'}(\ell)$, and every agent $i \in N$ picks his or her bundle from the budget defined by τ_{s_i} . Let b denote the lowest well-being level in the population at the optimal allocation.

Let us change the tax menu by simply replacing $\tau_{s_{\min}}$ with $\tau'_{s_{\min}}$, defined by: $\tau'_{s_{\min}}(\ell) = -b$, for all $\ell \in [0, 1]$. Observe that $\tau'_{s_{\min}}$ satisfies the requirement of the theorem. We claim that for all low-skilled agents, $c_i = s_{\min}\ell_i + b$, so z_N is still obtained with the new menu and the desired conclusion of the theorem is satisfied.

First, it is impossible that some low-skilled agent j has $c_j < s_{\min}\ell_j + b$ because this would imply that this agent's well-being level is less than b , contradicting the assumption that the lowest level is b . Now, suppose that there is a low-skilled agent j such that $c_j > s_{\min}\ell_j + b$. Under the new menu, there is a new allocation z'_N in which this agent “pays” $\tau'_{s_{\min}}(\ell'_j) = -b > s_{\min}\ell_j - c_j = \tau_{s_{\min}}(\ell_j)$. As $\tau'_{s_{\min}}(\ell'_i) = -b \geq \tau_{s_{\min}}(\ell_i)$ for all low-skilled i , and $z'_i = z_i$ for all higher-skilled agents, the new menu generates a budget surplus. At the new allocation z'_N the lowest well-being level is still b , and it is now attained by all low-skilled agents. One can redistribute the budget surplus of z'_N and obtain another feasible and incentive-compatible allocation z''_N that strictly Pareto-dominates z'_N . The lowest well-being level at z''_N is greater than b , which contradicts the assumption that z_N is optimal. This proves the claim of the previous paragraph.

The reasoning is identical if one looks at $\mathbf{R}^{\tilde{E}W}(E)$. If c denotes the lowest well-being level at the optimal allocation, then the new tax function $\tau'_{s_{\min}}$ one should look at is $\tau'_{s_{\min}}(\ell) = -(c - s_{\min}\tilde{\ell})$, for all $\ell \in [0, 1]$. Apart from that slight change, the proof is similar.

The reasoning is also similar for $\mathbf{R}^{\text{slex}}(E)$. If s denotes the lowest well-being level at the optimal allocation, the new tax function $\tau'_{s_{\min}}$ is defined by $\tau'_{s_{\min}}(\ell) = \max\{\tau_{s_{\min}}(\ell), \tau_{s_{\min}}(1)\}$.

Theorem 11.1 is a key result of this chapter. The axioms that justify $\mathbf{R}^{\text{sminlex}}$ and $\mathbf{R}^{\text{slex}}$, on the one hand, and $\mathbf{R}^{\tilde{E}W}$, on the other hand, are drastically different. The former SOFs satisfy equal-preferences transfer and, consequently, are definitely on the compensation side of the dilemma that we studied in the

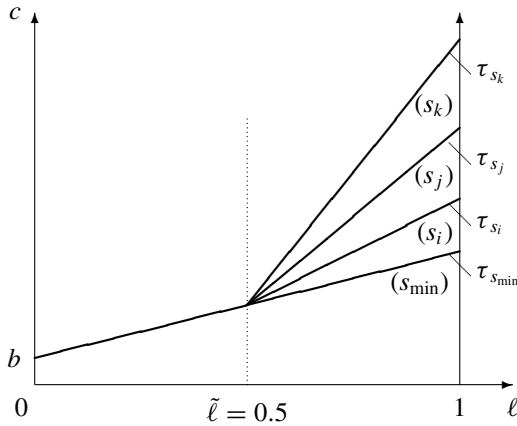


Figure 11.11. Optimal second-best allocations according to $\mathbf{R}^{\tilde{\ell}EW}$ when labor time is observable

previous chapter. The latter SOF satisfies equal-skill transfer and, consequently, is on the responsibility side of the dilemma. In spite of these opposing properties, second-best optimality, when labor time is observable, yields tax functions that exhibit a remarkable common feature: the marginal tax rate should be zero (or, on average, no greater than zero) on the earnings of low-skilled agents.

Under the assumption of observable labor time, $\mathbf{R}^{\tilde{\ell}EW}(E)$ turns out to be easier to study than the other SOFs. The next result is our first characterization of an optimal second-best allocation.

Theorem 11.2 *Assume labor time is observable. Let $E = (s_N, R_N) \in \mathcal{E}$ be such that for all $i \in N$, there is $j \in N$ such that $R_j = R_i$ and $s_j = s_{\min}$, and let $b \geq 0$. For each $s \in S$, let τ_s be defined by:*

$$\tau_s(\ell) = \begin{cases} (s - s_{\min})\ell - b & \text{if } \ell \leq \tilde{\ell} \\ (s - s_{\min})\tilde{\ell} - b & \text{if } \ell \geq \tilde{\ell}. \end{cases}$$

If an allocation obtained with the menu $\{\tau_s\}$ is second-best efficient, then it is second-best optimal for $\mathbf{R}^{\tilde{\ell}EW}(E)$.

Figure 11.11 represents a tax menu satisfying the definition given in the theorem. By construction, all low-skilled agents have an $\tilde{\ell}$ -equivalent well-being level of $b + s_{\min}\tilde{\ell}$. All higher-skilled agents working more than $\tilde{\ell}$ have the same well-being level, as their actual budget coincides with their implicit budget for all labor time weakly greater than $\tilde{\ell}$. The allocation generated by this menu of tax functions has the property that all agents working more than $\tilde{\ell}$ must be considered as equally bad off as the low-skilled agents. The only agents with a well-being level larger than $b + s_{\min}\tilde{\ell}$ are among the higher-skilled agents choosing to work less than $\tilde{\ell}$.

Assume that the allocation z_N generated by $\{\tau_s\}$ is second-best efficient. The intuition of the proof that it must be optimal is the following. Suppose that there exists another feasible and incentive-compatible allocation z'_N with a lowest well-being level that is greater than $b + s_{\min}\tilde{\ell}$. As z_N is second-best efficient, necessarily some agents are worse-off in z'_N . Given the observation we made in the previous paragraph, such agents cannot be low-skilled agents or agents choosing to work more than $\tilde{\ell}$, because their well-being level is the lowest at z_N . Consequently, any agent that is worse-off in the second allocation is a higher-skilled agent working less than $\tilde{\ell}$.

Here is where the assumption of the theorem about preferences enters into play. At least one low-skilled agent has the same preferences as that higher-skilled agent, makes the same choice (from a smaller budget) at z_N , and, consequently, must also be worse-off in z'_N than in z_N . This, however, contradicts the previous paragraph, and concludes the argument.

This result may seem to have limited scope because it is generally unlikely that the menu $\{\tau_s\}$, as defined in the proposition, generates a second-best efficient allocation. One can, however, safely conjecture that if the allocation obtained with this menu is not too inefficient, the optimal tax menu is close to $\{\tau_s\}$. For $\tilde{\ell} = 0$, this menu corresponds to the laissez-faire policy (one must then have $b = 0$), which yields an efficient allocation and is indeed optimal for $\mathbf{R}^{OEW}(E)$. The likelihood that the optimal menu is close to $\{\tau_s\}$ therefore increases when $\tilde{\ell}$ is smaller.

In practice, a menu such as $\{\tau_s\}$ is easy to enforce (assuming that labor time or wage rates are observable), and one can then proceed to check whether it generates large inefficiencies.

We now turn our attention to the second informational structure – that is, we now assume that labor time is no longer observable. When labor time is not observable, the optimal tax scheme is complex to analyze when there is no restriction on the distribution of skills and preferences. For the next results of this section, we need restrictions on the distribution of types in the economy. The intuitive meaning of these restrictions is that pretax income is not too informative a signal of one's type. More precisely, the idea is that over an interval of income $[0, s]$, it is impossible, by looking only at preferences restricted to that interval of income, to identify agents with greater productivity than s and distinguish them, on the basis of their preferences, from agents with productivity s .

The first assumption applies this idea to the interval $[0, s_{\min}]$ and requires that for any agent, there exists a low-skilled agent who has the same preferences in (y, c) space over the relevant range.

Assumption 11.1 *For all $i \in N$, there is $j \in N$ such that $s_j = s_{\min}$ and for all $(y, c), (y', c') \in [0, s_{\min}] \times \mathbb{R}_+$:*

$$\left(\frac{y}{s_j}, c\right) R_j \left(\frac{y'}{s_j}, c'\right) \Leftrightarrow \left(\frac{y}{s_i}, c\right) R_i \left(\frac{y'}{s_i}, c'\right).$$

The second restriction is more demanding and extends the idea to all relevant intervals. It requires that for any agent and any skill lower than that of this agent, there exists another agent with that lower skill who is indistinguishable in the relevant range. Let $\mathcal{S}(E)$ denote the set of skills in the current profile:

$$\mathcal{S}(E) = \{s \in S \mid \exists i \in N, s_i = s\}.$$

Assumption 11.2 *For all $i \in N$, all $s \in \mathcal{S}(E)$ such that $s < s_i$, there is $j \in N$ such that $s_j = s$ and for all $(y, c), (y', c') \in [0, s] \times \mathbb{R}_+$:*

$$\left(\frac{y}{s_j}, c\right) R_j \left(\frac{y'}{s_j}, c'\right) \Leftrightarrow \left(\frac{y}{s_i}, c\right) R_i \left(\frac{y'}{s_i}, c'\right).$$

These assumptions are a generalization of the restriction made in the previous section, in which agents of types 12 and 21 could not be distinguished. Let us insist that even if they are restrictive, we impose them only to exclude the implausible case in which fiscal authorities can design the tax for some interval $[y, y']$ (and, in particular, apply very high tax rates over that interval) knowing that only agents having an ability to earn much greater incomes would ever end up with an income in this interval.

Our first result is reminiscent of Theorem 11.1. Even when labor time is unobservable, both $\mathbf{R}^{s_{\min}, \text{lex}}$ and $\mathbf{R}^{\tilde{E}W}$ recommend to assign the same subsidy to all agents earning less than $s_{\min}$.

Theorem 11.3 *Assume labor time is unobservable. Let $E = (s_N, R_N) \in \mathcal{E}$ satisfy Assumption 11.1. Every second-best optimal allocation for $\mathbf{R}^{s_{\min}, \text{lex}}(E)$ or $\mathbf{R}^{\tilde{E}W}(E)$ can be obtained by a tax function τ that is constant over $[0, s_{\min}]$.*

The intuition for this result is similar to the intuition of Theorem 11.1. Let z_N be an optimal allocation with lowest well-being level b . Consider a tax function such as τ represented in Figure 11.12, constructed so the budget set is the lower envelope of the indifference curves of the population. In the figure, it does not satisfy the property stated in the theorem. By Assumption 11.1, the part of the budget set corresponding to $y \in [0, s_{\min}]$ is the lower envelope of the indifference curves of the low-skilled agents. Therefore b equals the lowest value of $-\tau(y)$ for $y \in [0, s_{\min}]$.

Now, consider the new tax function τ' and the resulting allocation z'_N . It consists of applying a constant amount of subsidy, b , to all income levels y such that $-\tau(y) \geq b$. In the figure, this affects all incomes lower than y' . Beyond y' , τ and τ' coincide. This rise in taxation keeps the minimal well-being level identical among agents earning less than $s_{\min}$, and it does not affect that of agents earning more than y' (those agents choose the same earnings under τ as under τ'). The agents earning between $s_{\min}$ and y' in z_N may be worse off in z'_N , but the lowest well-being level in z'_N , over the whole population, is still b .

If z_N can be obtained with τ' , the conclusion of the theorem is satisfied. If z_N cannot be obtained with τ' , this means that the tax paid by at least one agent has increased in z'_N . As, by construction, no agent has benefited from a tax reduction, the allocation generated by τ' generates a budget surplus. By

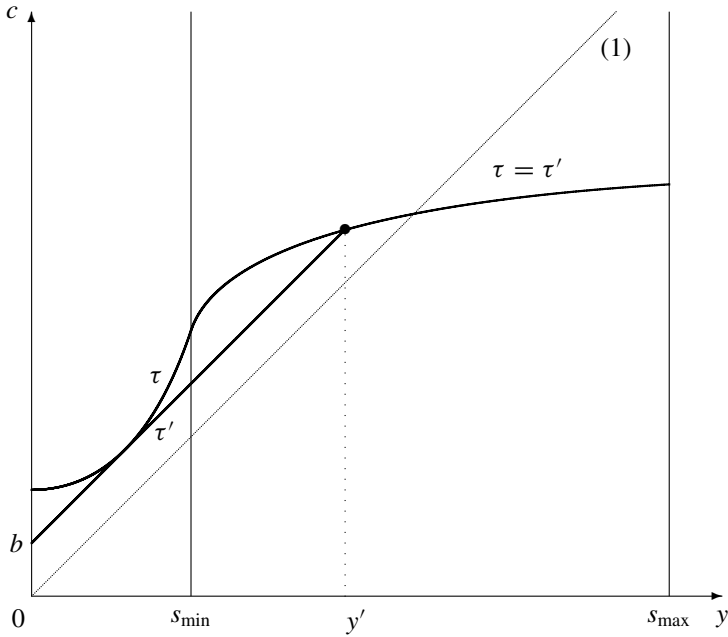


Figure 11.12. Optimal allocations for $\mathbf{R}^{s_{\min}\text{lex}}$ and $\mathbf{R}^{\tilde{E}W}$ have a zero marginal rate on low incomes.

redistributing that surplus to all agents (which can be done by slightly translating τ' upward), we can obtain a new allocation that strictly Pareto dominates the previous one, thereby strictly increasing its lowest well-being level above b , and contradicting the fact that the initial allocation z_N was optimal.

Our last three results are partial characterizations of the optimal allocations according to our three SOFs. Let us now examine the optimal allocations for $\mathbf{R}^{s_{\min}\text{lex}}$. The following theorem is almost a corollary of the previous one.

Theorem 11.4 *Assume labor time is unobservable. Let $E = (s_N, R_N) \in \mathcal{E}$ satisfy Assumption 11.1. A tax function maximizing $\mathbf{R}^{s_{\min}\text{lex}}(E)$ can be computed by minimizing $\tau(s_{\min})$ under the constraints that*

$$\tau(y) = \tau(s_{\min}), \text{ for all } y \leq s_{\min}$$

$$\tau(y) \geq \tau(s_{\min}), \text{ for all } y > s_{\min}.$$

The first constraint is a repetition of the zero marginal tax result of the previous theorem. The second constraint follows from the construction given earlier.

Theorem 11.4 is illustrated in Figure 11.13. In the earnings-consumption space, computing the optimal tax amounts to maximize the height of the point

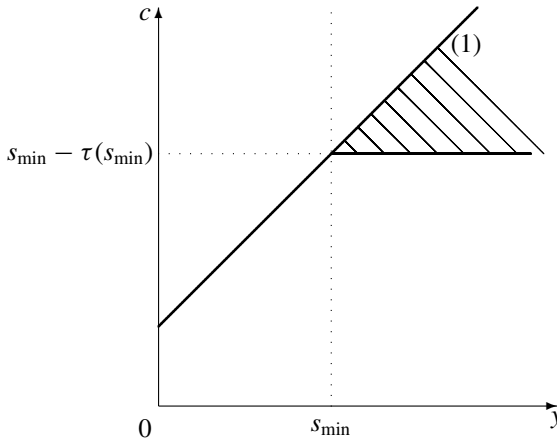


Figure 11.13. Optimal second-best allocations according to $\mathbf{R}^{s_{\min}^{\text{lex}}}$ when labor time is unobservable

$(s_{\min}, s_{\min} - \tau(s_{\min}))$ under the constraint that the income function $y - \tau(y)$ must be located in the area delineated by the thick lines.

Let us now turn to the study of our second SOF.

Theorem 11.5 *Assume labor time is unobservable. Let $E = (s_N, R_N) \in \mathcal{E}$ satisfy Assumption 11.1. A tax function maximizing $\mathbf{R}^{s_{\min}^{\text{lex}}}$ can be computed by maximizing the net income of the hardworking poor, $s_{\min} - \tau(s_{\min})$, under the constraints that*

$$\begin{aligned} \frac{\tau(y)}{y} &\leq \frac{\tau(s_{\min})}{s_{\min}} \quad \text{for all } y \in (0, s_{\min}], \\ \tau(y) &\geq \tau(s_{\min}) \quad \text{for all } y, \\ \tau(0) &\leq 0. \end{aligned}$$

The three constraints mean, respectively, that the average tax rate on low incomes is always lower than at $s_{\min}$, that the tax (subsidy) is the smallest (largest) at $s_{\min}$, and that the tax (subsidy) is nonpositive (nonnegative) at 0.

Theorem 11.5 is illustrated in Figure 11.14. From the point $(s_{\min}, s_{\min} - \tau(s_{\min}))$ one can construct the hatched area delimited by an upper line of slope 1 and a lower boundary made of the ray to the origin (on the left) and a flat line (on the right). Theorem 11.5 says that computing the optimal tax may, without welfare loss, be done by maximizing the second coordinate of the point $(s_{\min}, s_{\min} - \tau(s_{\min}))$ under the constraint that the income function $y - \tau(y)$ is located in the corresponding hatched area.

The intuition of this theorem parallels that of the previous one. First, the lower bound on the amount of taxes, or, as illustrated in the figure, the upper bound on the consumption function follows the same logic as the one illustrated in Figure 11.12: if the tax scheme does not satisfy this lower bound, then it is

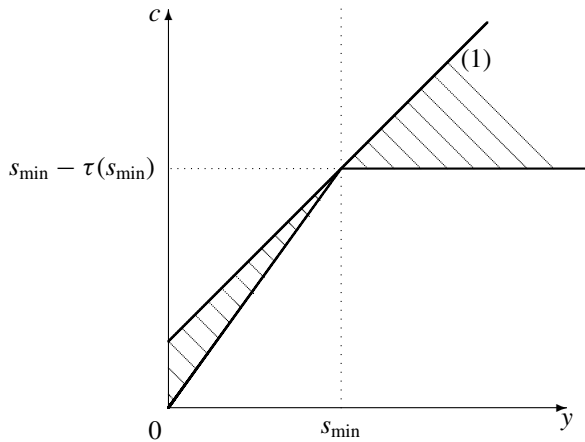


Figure 11.14. Optimal second-best allocations according to $\mathbf{R}^{\text{slex}}$ when labor time is unobservable

possible to construct another tax scheme that has this property and generates an allocation with the same associated lowest level of well-being.

Second, the upper bound on the amount of taxes, or the lower bound on the consumption function, follows from the criterion that should be maximized. As proved in the earlier section on reforms, we should try to maximize the average rate of subsidy among agents earning less than $s_{\min}$. That amounts to maximizing the consumption level of hardworking poor agents, under the constraint that no poorer agent gets a lower average rate of subsidy, as stated in the theorem.

This result does not say that every optimal tax must satisfy these constraints, but it says, quite relevantly for the social policymaker, that there is no problem – that is, no welfare loss – in restricting attention to taxes satisfying those constraints, when looking for the optimal allocation. This result shows how the social preferences defined in this chapter lead to focusing on the hardworking poor, who should get, in the optimal allocation, the greatest *absolute amount* of subsidy, among the whole population. However, the taxes computed for those with a lower income than $s_{\min}$ also matter, as those agents must obtain at least as great a *rate of subsidy* as the hardworking poor.

Our last result is a partial characterization of the optimal tax function according to $\mathbf{R}^{\tilde{E}W}$. Let $\tilde{\ell} \in [0, 1]$. Let us define the following tax function:

- (a) for all $y \in [0, s_{\min}]$, $\tau(y) = \tau(0) \leq 0$;
- (b) for all $s, s' \in S$ such that $s < s'$ and $s < s_i < s'$ for no $i \in N$, all $y \in [s, s']$,

$$\tau(y) = \min\{\tau(0) + (s - s_{\min})\tilde{\ell} + (y - s), \tau(0) + (s' - s_{\min})\tilde{\ell}\}.$$

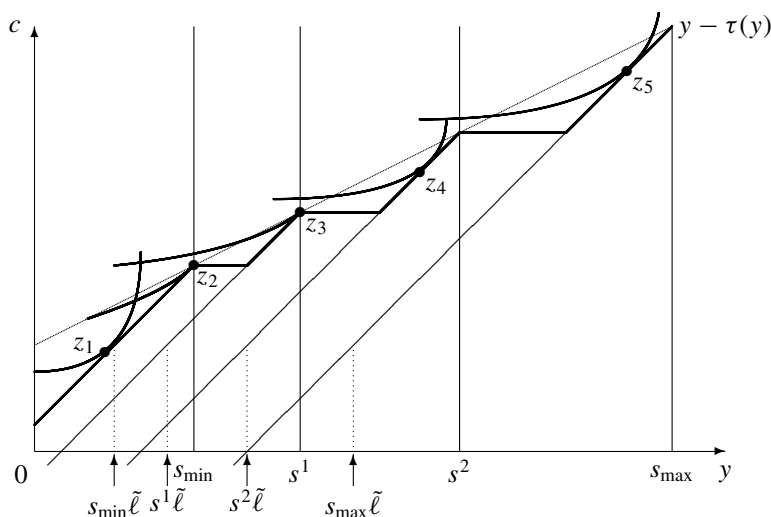


Figure 11.15. Optimal second-best allocations according to $\mathbf{R}^{\tilde{l}EW}$ when labor time is unobservable

A tax function of this kind will be called an $\tilde{l}$ -type tax. This formula calls for some explanation. The tax function is piecewise linear. The segment on low incomes $[0, s_{\min}]$ is constant, with a fixed subsidy $-\tau(0)$. Then comes a segment, $[s_{\min}, y^1]$, for some y^1 between $s_{\min}$ and the next element s^1 of $\mathcal{S}(E)$, where the rate of taxation is 100 percent. Of course, no individual is expected to earn an income in this interval. The next segment covers the interval $[y^1, s^1]$, and has a zero marginal tax rate. The function continues with successive pairs of intervals, one with 100 percent marginal tax and the other with a zero marginal tax. The key feature is that the points $(s, \tau(s))$, for $s \in \mathcal{S}(E)$, are aligned, and the slope of the line is precisely $\tilde{l}$; that is, for all $s, s' \in \mathcal{S}(E)$,

$$\frac{\tau(s) - \tau(s')}{s - s'} = \tilde{l}.$$

When $\mathcal{S}(E)$ is a large set with elements spread over the interval $[s_{\min}, s_{\max}]$, the tax function is therefore approximately a flat tax (constant marginal tax rate of $\tilde{l}$), except for the $[0, s_{\min}]$ interval, where it is constant.

The corresponding budget set delineated by $y - \tau(y)$ is illustrated in Figure 11.15, in which the indifference curves of five individuals are also depicted. Our last result follows.

Theorem 11.6 *Assume labor time is unobservable. Let $E = (s_N, R_N) \in \mathcal{E}$ satisfy Assumption 11.2. If an allocation obtained with an $\tilde{l}$ -type tax is second-best efficient, then it is second-best optimal for $\mathbf{R}^{\tilde{l}EW}(E)$.*

The proof proceeds in two steps. First, we identify the worst-off agents among agents of the same skill subgroup. As it is often the case, they will be those working full time, or, more generally, those having the largest labor time

in their subgroup. Second, we compare the well-being level of the agents who are the worst-off in their subgroup. The $\tilde{\ell}$ -type tax scheme is constructed in the precise way that equalizes the minimal well-being across skill subgroups. Let us go through each step in greater detail.

An interesting property of an $\tilde{\ell}$ -type tax function τ is that for all $i \in N$, if i chooses a bundle that is on the segment with slope 1 (zero marginal tax) just below s_i , then

$$u_{\tilde{\ell}}(z_i, s_i, R_i) = -\tau(0) + s_{\min}\tilde{\ell}.$$

Let us illustrate this fact in the case of individual 4, assuming that $s_4 = s^2$. From the second term in the definition of τ , one has

$$\tau(y_4) = \tau(0) + (s^2 - s_{\min})\tilde{\ell},$$

Bundle z_4 is optimal for R_4 in the budget set for which this level of tax is lump-sum. Therefore one simply has

$$\begin{aligned} u_{\tilde{\ell}}(z_4, s_4, R_4) &= -\tau(y_4) + s^2\tilde{\ell} \\ &= -\tau(0) + s_{\min}\tilde{\ell}, \end{aligned}$$

which achieves the proof.

If we assume that agent 4 has a larger skill than s^2 , such as $s_{\max}$, then agent 4's implicit budget is larger than that of agents with the same skill, such as agent 5. Agent 4, in that case, would have a larger implicit budget than agent 5, and, therefore, a larger well-being level. By Assumption 11.2, there exists an agent of skill s^2 who is undistinguishable from agent 4. The well-being of that agent is $-\tau(0) + s_{\min}\tilde{\ell}$. This proves that there necessarily exists an agent of skill s^2 with a well-being level of $-\tau(0) + s_{\min}\tilde{\ell}$, the lowest well-being level among agents of that skill.

The second step consists of checking that the quantity $-\tau(0) + s_{\min}\tilde{\ell}$ is independent of s_i . This tax function therefore equalizes $u_{\tilde{\ell}}$ across individuals who work full time or just below full time. This equality can be seen in Figure 11.15, in which the vertical segments at $s_{\min}\tilde{\ell}$, $s^1\tilde{\ell}$, $s^2\tilde{\ell}$, and $s_{\max}\tilde{\ell}$ between the horizontal axis and the lines of slope 1 corresponding to each of these values of s all have the same length.

Finally, suppose that there exists another feasible and incentive-compatible allocation z'_N with a greater minimal $u_{\tilde{\ell}}$. As z_N is second-best efficient, some agents must be worse-off in z'_N than in z_N . As z_N is optimal, such agents must have a greater well-being in z_N than $-\tau(0) + s_{\min}\tilde{\ell}$. However, by Assumption 11.2, for each agent of this sort there is another agent with lower skill and a well-being level in z_N at $-\tau(0) + s_{\min}\tilde{\ell}$. This was illustrated two paragraphs earlier with agent 4. In z'_N these agents must also be worse-off, falling below $-\tau(0) + s_{\min}\tilde{\ell}$. This contradiction proves that z_N is optimal.

Let us insist on the fact that an $\tilde{\ell}$ -type tax is likely to be optimal only if it is second-best efficient. It is clear, for instance, that the $\tilde{\ell}$ -type tax with $\tilde{\ell} = 1$ recommends a linear tax rate of 100 percent, which is obviously inefficient. The lower $\tilde{\ell}$, the more likely it is that the corresponding $\tilde{\ell}$ -type tax is optimal.

11.6 CONCLUSION

Chapters 10 and 11 together illustrate how the SOF approach developed in this book can be used to evaluate social policies. The first step consists of defining axioms capturing principles of fairness in the economic situations under scrutiny. There may be conflicts between these axioms, or between fairness and efficiency axioms. After the lists of compatible axioms are identified, cross-economy robustness axioms can be added to characterize precise objectives a policymaker should try to maximize. The second step deals with the implementation of these objectives when the policymaker faces informational constraints. In this chapter, incentive-compatibility constraints are such that we can restrict our attention to tax schemes. The criteria we eventually obtain, consequently, bear directly on the shape of the tax functions.

The applications in this chapter show that the only information one needs to evaluate and compare tax schemes are the relevant tax functions – that is, the relationship between labor time or earnings on one hand and consumption on the other. That information is easy to gather; it basically involves identifying the relevant tax legislation. As we have emphasized already, it is not necessary in such evaluation to know the distribution of skills and preferences. Policymakers, with this approach, obtain evaluation criteria that are readily applicable.

We made the assumption, in this chapter, that the minimal skill is strictly positive. In our model, the skill refers to the wage rate at which one is able to find a job. In situations of high unemployment, one can argue that the skill of some agents is actually zero. Under such an assumption, the criterion corresponding to either $\mathbf{R}^{\text{sllex}}$ or $\mathbf{R}^{\text{minlex}}$ boils down to maximizing the minimal consumption level. If the policymaker is interested in maximizing $\mathbf{R}^{\tilde{E}W}$, the zero marginal rate of taxation result disappears, but the objective is not necessarily to maximize the lowest consumption level. As the last theorem of this chapter may suggest, linear taxation (at a rate of $\tilde{\ell}$) is likely to be optimal.⁸

In almost all countries, one agent's labor income tax does depend on more than the agent's earnings. The typical additional argument is the size, or, more generally, the composition of the household. There may be other arguments as well, such as the working status of the person, or the person's age. The result of this chapter can still be applied in these cases, by considering each set of agents in the same categories in turn. The results presented here do not allow us, however, to draw conclusions about the relative tax agents should pay across different categories. This problem requires a richer model.

Our last comment is about the difference between leximin and maximin types of SOFs. In the previous chapters, we saw how the combination of efficiency, fairness, and robustness axioms forces us to adopt SOFs that maximize the minimal level of some well-being index, but it was frequently the case that, among all such SOFs, the leximin was impossible to single out axiomatically. Indeed, the results said only that an allocation with a greater lowest level of

⁸ See Fleurbaey and Maniquet (2011) for details.

well-being than another allocation was strictly better, remaining silent about pairs of allocations with the same lowest level of well-being. In this chapter, for the sake of simplicity, we always considered that the policymaker had a leximin type of SOF. We want to stress, now, that all the results we obtained in this chapter would be obtained with any kind of SOF displaying a strict preference for increasing the lowest well-being level, because this is the only element of social preferences that mattered in our proofs. In other words, the results we obtained in the axiomatic derivations of SOFs prove sufficient for second-best applications.